The formulas are exactly the strings with a formation sequence

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-syntax-and-semantics/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the inductive case argues the conjunction branch in full and closes the remaining six branches by parallel reasoning (source line 1328), the same display-then-parallel pattern as the author's prop:formed proof; the sister reference lem:fseq-init (source lines 1275-1288) is an initial-segment lemma of the same unit, not yet delivered, and is quoted only as a black box exactly as the author uses it.

Editorial note. The label thm:fseq-frm-equiv names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Every formula has a formation sequence. Same unit, prop:formed (delivered as L46214). Subset direction (proof lines 1299-1300): by prop:formed, Frm(L) is contained in the set of strings having a formation sequence; the theorem proves the converse.
- Initial segments of formation sequences. Same unit, lem:fseq-init (source lines 1275-1288). Inductive case (lines 1321-1323): initial segments (A_0,...,A_j) and (A_0,...,A_k) of the formation sequence are formation sequences for A_j and A_k.

Source reference. Original lemma lem-fseq-init: sources/olp-first-order-logic-syntax-and-semantics/pinned-original-source.tex, source lines 1276–1276. Prerequisite pointer invoked at proof line(s) 1321; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-formed: sources/olp-first-order-logic-syntax-and-semantics/pinned-original-source.tex, source lines 1230–1230.

Prerequisite pointer invoked at proof line(s) 1299; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1. $\mathrm{Frm}(\mathcal{L})$ *is the set of all* $\mathcal{L}L$*-strings* A *such that there exists a formula formation sequence for* A*.*

Proof. Let F be the set of all strings of symbols in the language $\mathcal{L}L$ that have a formation sequence. We have seen in prop-formed that $\mathrm{Frm}(\mathcal{L}) \subseteq F$, so now we prove the converse.

Suppose A has a formation sequence $\langle A_0, \ldots, A_n \rangle$. We prove that $A \in \mathrm{Frm}(\mathcal{L})$ by strong induction on n. Our induction hypothesis is that every string of symbols with a formation sequence of length $m < n$ is in $\mathrm{Frm}(\mathcal{L})$. By the definition of a formation sequence, either $A \equiv A_n$ is atomic or there must exist $j, k < n$ such that one of the following is the case:

1. $A \equiv \neg A_j$.
2. $A \equiv (A_j \wedge A_k)$.
3. $A \equiv (A_j \vee A_k)$.
4. $A \equiv (A_j \to A_k)$.
5. $A \equiv (A_j \leftrightarrow A_k)$.
6. $A \equiv \forall x\, A_j$.
7. $A \equiv \exists x\, A_j$.

Now we reason by cases. If A is atomic then $A_n \in \mathrm{Frm}(\mathcal{L}_0)$. Suppose instead that $A \equiv (A_j \wedge A_k)$. By lem-fseq-init, $\langle A_0, \ldots, A_j \rangle$ and $\langle A_0, \ldots, A_k \rangle$ are formation sequences for A_j and A_k, respectively. Since these are proper initial subsequences of the formation sequence for A, they both have length less than n. Therefore by the induction hypothesis, A_j and A_k are in $\mathrm{Frm}(\mathcal{L}_0)$, and by the definition of formula, so is $(A_j \wedge A_k)$. The other cases follow by parallel reasoning. $\square$